

Beyond the spine

The Spine harvests risk premia — validated, governed, live. This is the frontier behind it: a regime-aware statistical framework, *Gamma-ARMA*, that asks not what the market will do, but *which market we are in*. It is the thinking, not yet the product.

by **Carter Warrens** · research track · 2026 · companion to “Harvesting Risk, Not Return”

RESEARCH PREVIEW — NOT DEPLOYED

Nothing in this paper is running with capital or has cleared out-of-sample validation. What is validated and live today is the **Spine** (see the companion white paper). The figures here are *design targets, not results*; the validation is the explicit subject of the follow-up. We publish the thinking so you can see how it is done — and how carefully the door between “researched” and “deployed” is guarded.

ABSTRACT

Gamma-ARMA is a **regime-detection** framework. Instead of committing to one time-series model of the market, it maintains a probability distribution over *which* model is currently active — a Gamma distribution over ARMA/GARCH parameter space — and reads forward-looking, market-priced inputs (the volatility surface, the yield curve) that backward-looking estimation cannot supply. Its output is not a return forecast; it is a set of *risk parameters* — how much to hold, how to blend sleeves, how hard to brake — conditioned on the regime. It is the principled generalization of the crude drawdown brake the Spine already runs, and it will not touch capital until it clears the same validation the Spine did.

1 Why regime at all

Static models are quietly decaying. Passive flows are roughly half of US-equity assets; algorithmic execution and central-bank intervention increasingly set prices. Autocorrelation structures that held for decades are becoming less informative. The **2022 equity–bond correlation flip** —

negative to positive — invalidated twenty years of cross-asset relationships in a single regime change, and binary geopolitical events increasingly move markets with no autoregressive representation at all.

And at daily resolution the thing most models chase — *drift* — is almost pure noise:

$$\frac{|\mu_{\text{daily}}|}{\sigma_{\text{daily}}} \approx \frac{0.009\%}{1.5\%} \approx \frac{1}{170}$$

The daily drift of even a badly losing market is on the order of a hundredth of its daily volatility, so the sign of tomorrow is close to a coin flip (we measured exactly this — see the companion paper). The information lives in the *magnitude* distribution of moves — the volatility and tail structure — not in their direction. Model the noise, not the drift.

2 The idea: a distribution over models

Rather than fix an ARMA(p,q)+GARCH specification and hope it keeps describing the market, treat the model itself as uncertain. Maintain a posterior over its parameter space, with a Gamma prior that concentrates mass on the neighborhoods corresponding to distinct regimes — calm, stressed, crisis:

$$\underbrace{p(\theta \mid r_{1:t}, \mathcal{V}_t)}_{\text{which model is active}} \propto \underbrace{p(r_{1:t} \mid \theta)}_{\text{ARMA/GARCH fit}} \underbrace{\pi_{\Gamma}(\theta)}_{\text{Gamma regime prior}}$$

The likelihood is the ordinary fit to recent returns; the prior $\pi_{\Gamma}(\theta)$ encodes the regimes; and $\mathcal{V}_t$ are the forward-looking inputs — the volatility surface (VIX / VXV / VVIX / VIX1D, IV rank) and a smoothed yield curve — that carry information about structural uncertainty *before* it shows up in realized returns. The output is a regime probability, and from it a set of risk parameters. Nowhere is there a bet on direction.

3 Fat tails, taken seriously

For some asset tiers the tail index α sits below 2. That is not a nuisance to winsorize away — by the Generalized Central Limit Theorem it is **structural**: fat tails are self-similar under aggregation, cannot be diversified away within a tier, and Gaussian machinery will systematically misprice the tail. A framework that assigns a tail index *per regime* sizes risk against the distribution that is actually there rather than a convenient normal proxy. This is squarely a *risk* claim, not a return claim — which is why it belongs in the same family as the Spine.

4 Where it meets the validated core

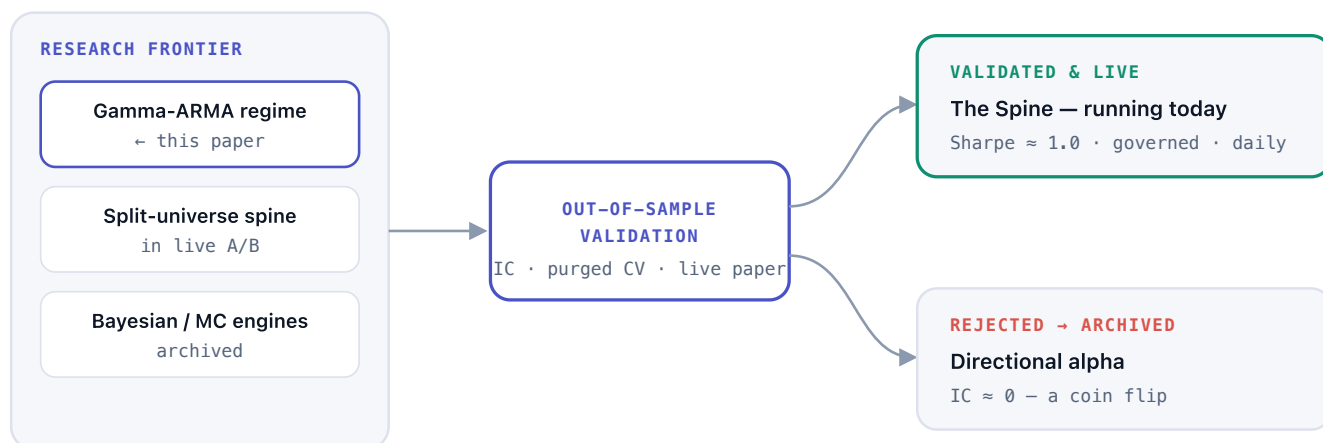
The Spine is not a rival to this work — it is its validated distillation. The Spine already carries a one-line regime model: a drawdown brake that halves gross exposure when the market is far below its recent peak. That brake is deliberately crude because crude is what survived validation. Gamma-ARMA is the principled generalization of that single line — a continuous regime estimate feeding the same risk-timing seam.

So the honest test is not a rewrite; it is an overlay. Drop the regime estimate into the Spine in place of the crude brake and measure it, out-of-sample, against what is already running. If the sophistication earns its keep, it graduates. If a one-line brake beats it, that is worth knowing too. Either way it is an experiment, not a claim.

5 The discipline is the product

What should reassure an allocator is not the machinery — it is what happens *to* the machinery. Everything here passes through one gate: out-of-sample validation. Only what clears it reaches capital. The Spine cleared it and runs today. A full directional-alpha engine — Bayesian filtering, Monte-Carlo, the works — did *not* (its information coefficient was indistinguishable from zero) and sits archived, preserved but unused. Gamma-ARMA is a candidate, not yet a survivor.

FIGURE — ONE GATE: RESEARCH IN, ONLY THE PROVEN OUT



Every candidate meets the same gate; only what clears it reaches capital. **We publish the rejections as readily as the survivors** — that discipline, not any single model, is the product.

6 What comes next

The follow-up paper *is* the validation: the regime engine tested as a Spine risk overlay, on the same honest terms, with the result — better, worse, or no different — reported either way. Alongside it, a broader pipeline is already moving: a split-universe spine now running in a live paper A/B, and further diversifying research behind it.

The Spine is what you can rely on today. This is a window into what is being built behind it — and the point of the window is not the view of the machinery, but the glass between it and your capital.

Validated strategy and full mathematics: [docs/whitepaper.html](#) and [docs/FINANCIAL_METHODS.md](#).

Architecture and research decisions: [docs/CANONICAL_ARCHITECTURE.md](#). Reference implementation (Julia): [the Blaque Baux repository](#).

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